ORBIT-PK Sample

600 subjects, 100 from each drug/route/arm group, sliced bit-identical from the full ORBIT-PK Dataset — not a re-simulation.

600	6	7,200	Free
Subjects	Groups	Observation rows	Price

SAMPLE OF	EDITION	FORMAT
ORBIT-PK Dataset	2.0	Apache Parquet, 6 files
SUBJECTS	**DRUGS**	**CONSTRUCTION**
600 (100 × 6 groups)	acetaminophen, scopolamine, zolpidem	bit-identical slice, verified
FULL DATASET PRICE	**THIS SAMPLE**	**LITERATURE BASE**
$600 USD	free	42 cited sources (shared)

“Every row in this sample is a real cell from the full dataset's own Parquet files, never a recomputed stand-in.”

This is a fully synthetic sample. No real crew member's or patient's data appears anywhere in this delivery, here or in the full dataset it is sliced from.

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01 · WHAT THIS SAMPLE IS

What This Sample Is

This is the free sample of the ORBIT-PK Dataset: 600 simulated adult subjects, 100 from each of the dataset's six drug/route/arm groups (acetaminophen oral GROUND and FLIGHT, scopolamine oral GROUND and FLIGHT, scopolamine transdermal-patch GROUND, zolpidem oral GROUND). Every value in it is sliced directly out of the real, already-generated main dataset files — not re-simulated, not re-rounded, not regenerated with a different random seed. Section 05 shows exactly how that bit-identical slice was verified.

The point of this sample is to let you check column shapes, value ranges, and the reasoning behind the model against real rows before paying for the full 105,000-subject dataset. If a number in the Data Passport looks surprising, you can open these files and see it for yourself at 1/175th scale.

02 · WHAT'S INSIDE — THE SIX FILES

What's Inside — the Six Files

Six files, the same shapes as the six tables in the full dataset, joined by the same `subject_id` key.

SAMPLE FILES 600 subjects, 6 files

File	Rows	What it is
subjects_sample.parquet	600	One row per sampled subject: drug, route, arm, dose, body weight, age.
truth_pk_parameters_sample.parquet	600	Exact, noise-free PK parameters and the microgravity perturbation actually applied, for these 600 subjects.
truth_pk_metrics_sample.parquet	600	Exact, noise-free Cmax, Tmax, AUC, elimination rate, and half-life for these 600 subjects.
observations_sample.parquet	7,200	Twelve noisy blood-concentration points per subject (600 × 12).
outcomes_sample.parquet	600	Each sampled subject's Cmax/AUC compared against the reference interval from the full dataset's own GROUND arm.
zolpidem_multidose_steady_state_sample.parquet	100	Steady-state Cmax/Cmin/AUC and accumulation ratios for the 100 sampled zolpidem subjects.

The 600 subjects split evenly across the dataset's six drug/route/arm groups, 100 each — not a random 600 out of 105,000, but a stratified 100-per-group slice, so every group is represented even though zolpidem's full GROUND group alone has 30,000 subjects.

SAMPLED GROUPS 6 groups, 100 each, 600 total

Drug	Route	Arm	Subjects
Acetaminophen	ORAL	GROUND	100
Acetaminophen	ORAL	FLIGHT	100
Scopolamine	ORAL	GROUND	100
Scopolamine	ORAL	FLIGHT	100
Scopolamine	PATCH (transdermal)	GROUND	100
Zolpidem	ORAL	GROUND	100

A `schema.csv` and the same plain-language Parquet guide as the full dataset ship with the sample. The schema is reproduced in Section 06.

03 · HONEST LIMITATIONS

Honest Limitations

THIS SAMPLE IS TOO SMALL FOR POPULATION-LEVEL STATISTICS

100 subjects per group is enough to check column shapes, value ranges, and a few individual rows against the reasoning in the Data Passport. It is not enough to re-derive medians or reference intervals that match the full dataset exactly — those are computed in the full dataset from the complete 15,000–30,000-subject groups, not from this 600-subject slice. `outcomes_sample.parquet` 's reference-interval bounds come from the full dataset's GROUND arm, not from the sample's own 100 GROUND subjects.

SAME SCOPE GAPS AS THE FULL DATASET

No FLIGHT arm exists for zolpidem or for the scopolamine PATCH route, here or in the full dataset — a documented negative search result, not something fixed by a larger sample. Both routes ship GROUND-only for the reasons given in the Data Passport.

NOT A STATISTICALLY REPRESENTATIVE RE-SAMPLE OF EACH GROUP

The 100 subjects per group are the first 100 rows of that group as the full dataset generator wrote it, not a random draw. That is enough to see the real shape of the distributions, but anyone building their own statistical test against this sample specifically (rather than against the full dataset) should be aware the 100 rows are not an independent random sample of the full 15,000–30,000-subject group.

04 · GET THE FULL DATASET

Get the Full Dataset

$600 USD

One-time purchase. All six full tables (105,000 subjects total), the full schema, the Data Passport, and the Validation Report.

The full dataset carries the same 42 cited sources, the same 41/41 validation checks, and the same two disclosed FAIL_DISCLOSED benchmark results as this sample — nothing about the model's honesty bar changes with scale. What scales up is only the number of subjects per group, from 100 to 15,000 or 30,000, and the observation table, from 7,200 rows to 1,260,000.

A companion **ORBIT-PK Generator** ($1,590) ships the full simulation source code separately, for anyone who needs subjects, doses, or random seeds beyond this fixed delivery. Both the dataset and the generator are listed on ai.market; checkout and licensing happen there.

05 · HOW THIS SAMPLE WAS BUILT — BIT-IDENTICAL BY CONSTRUCTION

How This Sample Was Built — Bit-Identical by Construction

This sample is not a second simulation run with a smaller subject count. It is sliced directly out of the full dataset's own already-generated Parquet files, using the same `subject_id` values the main generator assigned. For each of the six drug/route/ arm groups, the first 100 subject IDs belonging to that group are selected from `subjects.parquet`, and every other table (`truth_pk_parameters`, `truth_pk_metrics`, `observations`, `outcomes`, and, for zolpidem subjects, `zolpidem_multidose_steady_state`) is filtered to exactly that same set of IDs. There is no separate code path that recomputes a single number.

The build script (`BUILD/build_sample.py`) checks this directly, not just by construction: after writing the sliced files, it reads `truth_pk_parameters` back in from both the sample output and the main dataset, restricts both to the sampled IDs, sorts by `subject_id`, and asserts the two are exactly `.equals()` in pandas — a real equality check, not a shape check. That assertion passed when the sample was built, and a fresh check of the delivered files on 2026-10-29 confirms this sample's 600 subjects are bit-identical to the same 600 subjects' rows in the full dataset.

06 · SCHEMA

Schema

These six files use the exact same column names, types, and units as the full dataset's six tables (reproduced in full in the Data Passport's own schema section) — only the row counts differ. The table below groups closely related columns per file for a compact overview; `schema.csv` in this delivery lists every column on its own row.

subjects_sample.parquet

Column(s)	Type	Description	Origin
subject_id	str	Join key, same values as the full dataset's subject_id for these 600 subjects.	key
drug	str	One of acetaminophen, scopolamine, zolpidem.	label
arm	str	GROUND or FLIGHT. Zolpidem and the PATCH route have no FLIGHT rows, here or in the full dataset.	label
route	str	ORAL or PATCH (transdermal, scopolamine only).	label
dose_mg	float64 (mg)	Single oral dose, or total patch-delivered dose over the wear period.	label
body_weight_kg	float64 (kg)	Individually drawn for acetaminophen/scopolamine; flat 70 kg for zolpidem (disclosed).	model output
age_years	float64 (years)	Populated for acetaminophen only; NULL for scopolamine and zolpidem.	model output

truth_pk_parameters_sample.parquet

Column(s)	Type	Description	Origin
ka_true, F_true, CL_true, V1_true, Q_true, V2_true	float64	This subject's TRUE PK parameters after variability, covariates, and any flight perturbation. ka_true/F_true are NULL for PATCH subjects.	model output
f_ka_applied, f_F_applied, f_CL_applied	float64 (ratio)	Microgravity perturbation factors actually applied (exactly 1.0 for GROUND subjects).	derived
perturbation_source_study, mission_duration_class	str	Which real study/mission class an acetaminophen FLIGHT subject's perturbation came from; N/A labels otherwise.	label

truth_pk_metrics_sample.parquet

Column(s)	Type	Description	Origin
Cmax_true, Tmax_true, AUC_true, lambda_z_true, t_half_true	float64	This subject's TRUE peak concentration, time to peak, exposure, elimination rate, and half-life.	model output / derived

observations_sample.parquet

Column(s)	Type	Description	Origin
sample_time_h	float64 (hour)	Nominal blood-draw time: 0.25–24h for ORAL, 2–96h for PATCH.	scan design

Column(s)	Type	Description	Origin
concentration_observed	float64 (mg/L)	Simulated observed concentration with drug-appropriate noise added.	derived

outcomes_sample.parquet

Column(s)	Type	Description	Origin
cmax_reference_lo/hi, auc_reference_lo/hi	float64	Reference-interval bounds, computed from the FULL dataset's own GROUND arm (not from this sample's 100 GROUND subjects).	derived
cmax_outcome, auc_outcome	str	BELOW/WITHIN/ABOVE_REFERENCE_RANGE. Descriptive only, not a clinical classification.	derived

zolpidem_multidose_steady_state_sample.parquet

Column(s)	Type	Description	Origin
Cmax_steady_state, Cmin_steady_state, AUC_tau_steady_state	float64	This subject's TRUE steady-state metrics after 7 nightly doses.	model output
accumulation_ratio_observed, accumulation_ratio_theoretical	float64 (ratio)	Simulated and theoretical accumulation ratios; both close to 1.0 given zolpidem's short half-life.	derived

07 · REFERENCE LIST

Reference List

This sample shares its model and every cited source with the full ORBIT-PK Dataset. All 42 sources are listed here in full, in the order first introduced in the Data Passport, so this document stands on its own without requiring the full dataset's documentation to verify a citation.

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